

Introduction to Abstract Algebra III: Optional Homework

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Exercise 39.33. Let E be an extension field of a field F and let $\alpha \in E$ be transcendental over F . Show that every element of $F(\alpha)$ that is not in F is also transcendental over F .

Solution. Since α is transcendental over F , the field $F(\alpha)$ is isomorphic to the field of rational functions $F(x)$, so every element of $F(\alpha)$ has the form

$$\beta = \frac{f(\alpha)}{g(\alpha)},$$

where $f, g \in F[x]$ with $g \neq 0$. Suppose $\beta \in F(\alpha) \setminus F$; we wish to show β is transcendental over F .

Since $\beta \notin F$, the rational function $f(x)/g(x)$ is non-constant, so $h(x) := f(x) - \beta g(x) \in F(\beta)[x]$ is a nonzero polynomial (in x) of degree $n = \max(\deg f, \deg g) \geq 1$. Moreover, α is a root of h , since

$$h(\alpha) = f(\alpha) - \beta g(\alpha) = f(\alpha) - \frac{f(\alpha)}{g(\alpha)} \cdot g(\alpha) = 0.$$

Hence α is algebraic over $F(\beta)$.

Now suppose for contradiction that β is algebraic over F . Then $F(\beta)/F$ is a finite algebraic extension, and since α is algebraic over $F(\beta)$, the tower

$$F \subseteq F(\beta) \subseteq F(\beta, \alpha) = F(\alpha),$$

consists of algebraic extensions, so α is algebraic over F , a contradiction. Therefore β is transcendental over F . $\square$

Exercise 39.35. Following the idea of Exercise 31, show that there exists a field of 8 elements; of 16 elements; of 25 elements.

Solution. For a field of 8 elements, consider $p(x) = x^3 + x + 1 \in \mathbb{Z}_2[x]$. Since $p(0) = 1 \neq 0$ and $p(1) = 1 \neq 0$ in $\mathbb{Z}_2$, $p(x)$ has no roots in $\mathbb{Z}_2$ and is therefore irreducible over $\mathbb{Z}_2$ (as it has degree 3). Hence $\mathbb{Z}_2[x]/\langle x^3 + x + 1 \rangle$ is a field. Its elements are the cosets of polynomials of degree less than 3 over $\mathbb{Z}_2$, of which there are $2^3 = 8$.

Next, for a field of 16 elements, consider $p(x) = x^4 + x + 1 \in \mathbb{Z}_2[x]$. Since $p(0) = 1 \neq 0$ and $p(1) = 1 \neq 0$ in $\mathbb{Z}_2$, $p(x)$ has no roots in $\mathbb{Z}_2$, so $p(x)$ has no linear factors. One checks that $p(x)$ is not divisible by the only irreducible quadratic over $\mathbb{Z}_2$, namely $x^2 + x + 1$: performing the division gives remainder x , which is nonzero. Hence $p(x)$ is irreducible over $\mathbb{Z}_2$, and $\mathbb{Z}_2[x]/\langle x^4 + x + 1 \rangle$ is a field with $2^4 = 16$ elements.

Lastly, for a field of 25 elements, consider $p(x) = x^2 + x + 1 \in \mathbb{Z}_5[x]$. Evaluating at each element of $\mathbb{Z}_5$:

$$p(0) = 1, \quad p(1) = 3, \quad p(2) = 7 = 2, \quad p(3) = 13 = 3, \quad p(4) = 21 = 1,$$

so $p(x)$ has no roots in $\mathbb{Z}_5$ and is therefore irreducible over $\mathbb{Z}_5$ (as it has degree 2). Hence $\mathbb{Z}_5[x]/\langle x^2 + x + 1 \rangle$ is a field with $5^2 = 25$ elements. $\square$

Exercise 39.36. Let F be a finite field of characteristic p . Show that every element of F is algebraic over the prime field $\mathbb{Z}_p \leq F$. [Hint: Let F^* be the set of nonzero elements of F . Apply group theory to the group $\langle F^*, \cdot \rangle$ to show that every $\alpha \in F^*$ is a zero of some polynomial in $\mathbb{Z}_p[x]$ of the form $x^n - 1$.]

Solution. Since F has characteristic p , its prime subfield is $\mathbb{Z}_p$. The zero element $0 \in F$ is a root of $x \in \mathbb{Z}_p[x]$, hence algebraic over $\mathbb{Z}_p$. It remains to show every $\alpha \in F^*$ is algebraic over $\mathbb{Z}_p$.

Since F is finite, so is $F^* = F \setminus \{0\}$, and $\langle F^*, \cdot \rangle$ is a finite group of some order $n = |F^*| = |F| - 1$. By Lagrange's theorem, the order of every element divides $|F^*|$, so $\alpha^n = 1$ for every $\alpha \in F^*$. Hence every $\alpha \in F^*$ is a root of

$$x^n - 1 \in \mathbb{Z}_p[x],$$

and is therefore algebraic over $\mathbb{Z}_p$. Thus every element of F is algebraic over $\mathbb{Z}_p$. $\square$

Exercise 40.26. Let E be a finite extension field of F . Let D be an integral domain such that $F \subseteq D \subseteq E$. Show that D is a field.

Solution. Let $\alpha \in D$ be nonzero; we must show $\alpha^{-1} \in D$. Since $F \subseteq D \subseteq E$ and E is a finite extension of F , the element $\alpha \in E$ is algebraic over F . Hence α is also algebraic over D (as $F \subseteq D$), but more directly: since $[E : F]$ is finite, α is algebraic over F , so consider the map

$$\varphi : D \rightarrow D \quad \text{defined by} \quad \varphi(\beta) = \alpha\beta.$$

This is an F -linear map on D , viewed as a subspace of E . Since $[E : F]$ is finite, D is a finite-dimensional vector space over F , and φ is injective (as D is an integral domain and $\alpha \neq 0$). A linear map from a finite-dimensional vector space to itself is injective if and only if it is surjective, so φ is surjective. In particular, there exists $\beta \in D$ such that $\alpha\beta = 1$, i.e., $\alpha^{-1} = \beta \in D$. Since $\alpha \in D$ was an arbitrary nonzero element, D is a field. $\square$

Exercise 40.31. Show that if F , E , and K are fields with $F \leq E \leq K$, then K is algebraic over F if and only if E is algebraic over F , and K is algebraic over E . (You must not assume the extensions are finite.)

Solution. Suppose K is algebraic over F . Since $E \subseteq K$, every element of E is an element of K and hence algebraic over F , so E is algebraic over F . Now let $\alpha \in K$; since α is algebraic over F and $F \subseteq E$, any polynomial in $F[x]$ witnessing this lies in $E[x]$ as well, so α is algebraic over E . Hence K is algebraic over E .

Conversely, suppose E is algebraic over F and K is algebraic over E . Let $\alpha \in K$; we show α is algebraic over F . Since α is algebraic over E , there exists a nonzero polynomial

$$p(x) = a_n x^n + \cdots + a_1 x + a_0 \in E[x],$$

such that $p(\alpha) = 0$. The coefficients $a_0, a_1, \dots, a_n$ are elements of E , each algebraic over F . Setting $F_0 = F$ and $F_k = F(a_0, \dots, a_{k-1})$ for $k = 1, \dots, n+1$, each extension $F_k \leq F_{k+1}$ is finite since a_k is algebraic over $F \subseteq F_k$, and so by the tower theorem

$$[F_{n+1} : F] = [F_{n+1} : F_n] \cdots [F_1 : F] < \infty.$$

Since $p(x) \in F_{n+1}[x]$ and $p(\alpha) = 0$, the element α is algebraic over F_{n+1} , so $[F_{n+1}(\alpha) : F_{n+1}]$ is finite. By the tower theorem,

$$[F_{n+1}(\alpha) : F] = [F_{n+1}(\alpha) : F_{n+1}][F_{n+1} : F] < \infty,$$

so α is algebraic over F . Since $\alpha \in K$ was arbitrary, K is algebraic over F . $\square$

Exercise 40.32. Let E be an extension field of a field F . Prove that every $\alpha \in E$ that is not in the algebraic closure $\bar{F}_E$ of F in E is transcendental over $\bar{F}_E$.

Solution. Recall that $\bar{F}_E = \{\alpha \in E \mid \alpha \text{ is algebraic over } F\}$ is the algebraic closure of F in E , and is itself a field. Let $\alpha \in E \setminus \bar{F}_E$, so that α is transcendental over F . Suppose for contradiction that α is algebraic over $\bar{F}_E$. Then by Problem 40.31 (with the tower $F \leq \bar{F}_E \leq \bar{F}_E(\alpha)$), since $\bar{F}_E$ is algebraic over F and α is algebraic over $\bar{F}_E$, it follows that α is algebraic over F . This means $\alpha \in \bar{F}_E$, a contradiction. Therefore α is transcendental over $\bar{F}_E$. $\square$

Exercise 40.34. Show that if E is an algebraic extension of a field F and contains all zeros in F of every $f(x) \in F[x]$, then E is an algebraically closed field.

Solution. To show E is algebraically closed, we must show every nonconstant polynomial $p(x) \in E[x]$ has a zero in E . Let $p(x) \in E[x]$ be nonconstant, and let α be a zero of $p(x)$ in some algebraic closure of E . Since α is algebraic over E and E is algebraic over F , Problem 40.31 gives that α is algebraic over F , so $\alpha \in \bar{F}$.

Since α is algebraic over F , it has a minimal polynomial $m(x) \in F[x]$. By hypothesis, E contains all zeros in $\bar{F}$ of every polynomial in $F[x]$, so in particular E contains all zeros of $m(x)$. Since α is one such zero, we conclude $\alpha \in E$.

Thus every nonconstant polynomial in $E[x]$ has a zero in E , so E is algebraically closed. $\square$

Exercise 40.37. Argue that every finite extension field of $\mathbb{R}$ is either $\mathbb{R}$ itself or is isomorphic to $\mathbb{C}$.

Solution. Let E be a finite extension of $\mathbb{R}$. Since $[E : \mathbb{R}]$ is finite, E is an algebraic extension of $\mathbb{R}$, so every $\alpha \in E$ is algebraic over $\mathbb{R}$. Since $\mathbb{C}$ is algebraically closed (by the Fundamental Theorem of Algebra), the algebraic closure of $\mathbb{R}$ is $\mathbb{C}$, and every element of E lies in $\mathbb{C}$. Thus $[E : \mathbb{R}]$ divides $[\mathbb{C} : \mathbb{R}] = 2$, so either $[E : \mathbb{R}] = 1$ or $[E : \mathbb{R}] = 2$.

If $[E : \mathbb{R}] = 1$, then $E = \mathbb{R}$. If $[E : \mathbb{R}] = 2$, then E contains some $\alpha \notin \mathbb{R}$, which is algebraic over $\mathbb{R}$ with a minimal polynomial of degree 2. Since $\mathbb{R}$ has no irreducible polynomials of degree 2 with real roots, this minimal polynomial is an irreducible quadratic, and completing the square shows $\alpha = a + bi$ for some $a \in \mathbb{R}$ and $b \in \mathbb{R} \setminus \{0\}$. Hence $E = \mathbb{R}(\alpha) \cong \mathbb{R}[x]/\langle m(x) \rangle \cong \mathbb{C}$. $\square$

Exercise 40.38. Use Zorn's lemma to show that every proper ideal of a ring R with unity is contained in some maximal ideal.

Solution. Let N be a proper ideal of R . Define

$$\mathcal{S} = \{I \mid I \text{ is a proper ideal of } R \text{ and } N \subseteq I\},$$

partially ordered by inclusion. Since $N \in \mathcal{S}$, we have $\mathcal{S} \neq \emptyset$. Let $\mathcal{C} = \{I_\lambda\}_{\lambda \in \Lambda}$ be a chain in $\mathcal{S}$, and set $U = \bigcup_{\lambda \in \Lambda} I_\lambda$. We verify that U is a proper ideal of R containing N :

- $N \subseteq I_\lambda \subseteq U$ for all λ , so $N \subseteq U$.
- If $a, b \in U$, then $a \in I_\lambda$ and $b \in I_\mu$ for some λ, μ . Since $\mathcal{C}$ is a chain, one contains the other, say $I_\lambda \subseteq I_\mu$, so $a, b \in I_\mu$ and thus $a - b \in I_\mu \subseteq U$.
- If $a \in U$ and $r \in R$, then $a \in I_\lambda$ for some λ , so $ra, ar \in I_\lambda \subseteq U$.
- Since $1 \notin I_\lambda$ for any λ (each I_λ is proper), we have $1 \notin U$, so $U \neq R$.

Thus $U \in \mathcal{S}$ is an upper bound for $\mathcal{C}$. Since every chain in $\mathcal{S}$ has an upper bound in $\mathcal{S}$, Zorn's lemma guarantees a maximal element $M \in \mathcal{S}$. By definition M is a proper ideal containing N , and maximality in $\mathcal{S}$ means no proper ideal of R properly contains M , so M is a maximal ideal of R . $\square$

Exercise 42.15. Let F be an arbitrary field. We define the *derivative* of $p(x) = \sum_{k=0}^n a_k x^k \in F[x]$ to be $D(p(x)) = \sum_{k=0}^n (k \cdot (a_k x^{k-1}))$. Let $p(x), q(x) \in F[x]$, and let n and m be nonnegative integers. Prove the following statements.

- (i) $D(p(x) + q(x)) = D(p(x)) + D(q(x))$.
- (ii) $D(ap(x)) = aD(p(x))$ for any $a \in F$.
- (iii) $D(x^n x^m) = x^n D(x^m) + D(x^n) x^m$.
- (iv) $D(p(x) x^m) = p(x) D(x^m) + D(p(x)) x^m$.
- (v) $D(p(x) q(x)) = p(x) D(q(x)) + D(p(x)) q(x)$.

(vi) $(x - a)^2$ divides $p(x)$ if and only if a is a zero of both $p(x)$ and $D(p(x))$.

(vii) Give a proof of Lemma 42.8 using part (f).

Solution to (i). Let $p(x) = \sum_{k=0}^n a_k x^k$ and $q(x) = \sum_{k=0}^m b_k x^k$. Without loss of generality assume $n \leq m$ and set $a_k = 0$ for $k > n$. Then

$$D(p(x)+q(x)) = D\left(\sum_{k=0}^m (a_k + b_k)x^k\right) = \sum_{k=0}^m k(a_k + b_k)x^{k-1} = \sum_{k=0}^m k a_k x^{k-1} + \sum_{k=0}^m k b_k x^{k-1} = D(p(x)) + D(q(x)).$$

□

Solution to (ii). Let $p(x) = \sum_{k=0}^n a_k x^k$ and $a \in F$. Then

$$D(ap(x)) = D\left(\sum_{k=0}^n a a_k x^k\right) = \sum_{k=0}^n k(a a_k)x^{k-1} = a \sum_{k=0}^n k a_k x^{k-1} = a D(p(x)).$$

□

Solution to (iii). Using $x^n x^m = x^{n+m}$ and the definition of D :

$$D(x^n x^m) = D(x^{n+m}) = (n+m)x^{n+m-1} = n x^{n-1} x^m + x^n m x^{m-1} = D(x^n)x^m + x^n D(x^m).$$

□

Solution to (iv). Write $p(x) = \sum_{k=0}^n a_k x^k$. Then by parts (i), (ii), and (iii):

$$\begin{aligned} D(p(x)x^m) &= D\left(\sum_{k=0}^n a_k x^{k+m}\right) \\ &= \sum_{k=0}^n a_k D(x^{k+m}) = \sum_{k=0}^n a_k (x^k D(x^m) + D(x^k)x^m) \\ &= \left(\sum_{k=0}^n a_k x^k\right) D(x^m) + \left(\sum_{k=0}^n a_k D(x^k)\right) x^m \\ &= p(x)D(x^m) + D(p(x))x^m. \end{aligned}$$

□

Solution to (v). Write $q(x) = \sum_{j=0}^m b_j x^j$. Then by parts (i), (ii), and (iv):

$$\begin{aligned} D(p(x)q(x)) &= D\left(\sum_{j=0}^m b_j p(x)x^j\right) = \sum_{j=0}^m b_j D(p(x)x^j) \\ &= \sum_{j=0}^m b_j (p(x)D(x^j) + D(p(x))x^j) \\ &= p(x) \sum_{j=0}^m b_j D(x^j) + D(p(x)) \sum_{j=0}^m b_j x^j \\ &= p(x)D(q(x)) + D(p(x))q(x). \end{aligned}$$

□

Solution to (vi). Suppose $(x - a)^2 \mid p(x)$, so $p(x) = (x - a)^2 q(x)$ for some $q(x) \in F[x]$. Then clearly $p(a) = 0$. By part (v),

$$D(p(x)) = (x - a)^2 D(q(x)) + D((x - a)^2)q(x) = (x - a)^2 D(q(x)) + 2(x - a)q(x),$$

so $D(p(a)) = 0$. Hence a is a zero of both $p(x)$ and $D(p(x))$.

Conversely, suppose $p(a) = 0$ and $D(p(a)) = 0$. Since a is a zero of $p(x)$, we have $p(x) = (x - a)q(x)$ for some $q(x) \in F[x]$. By part (v),

$$D(p(x)) = (x - a)D(q(x)) + q(x).$$

Evaluating at a : $0 = D(p(a)) = q(a)$, so $(x - a) \mid q(x)$, say $q(x) = (x - a)r(x)$. Therefore $p(x) = (x - a)^2 r(x)$, so $(x - a)^2 \mid p(x)$. □

Solution to (vii). We prove Lemma 42.8: if F has prime characteristic p with algebraic closure $\bar{F}$, then $f(x) = x^{p^n} - x$ has p^n distinct zeros in $\bar{F}$.

Since $\bar{F}$ is algebraically closed, $f(x)$ splits completely over $\bar{F}$ and has at most p^n zeros. To show it has exactly p^n distinct zeros, it suffices by part (vi) to show $f(x)$ and $D(f(x))$ share no common zeros. Computing the derivative:

$$D(f(x)) = D(x^{p^n} - x) = p^n x^{p^n-1} - 1.$$

Since F has characteristic p , we have $p^n \equiv 0$ in F , so $D(f(x)) = -1$. This is a nonzero constant with no zeros in $\bar{F}$, so $f(x)$ and $D(f(x))$ share no common zeros. By part (vi), $(x - a)^2 \nmid f(x)$ for any $a \in \bar{F}$, meaning all zeros of $f(x)$ are simple. Hence $f(x)$ has p^n distinct zeros in $\bar{F}$. $\square$